20 Implement Future Sales Prediction using a suitable machine learning algorithm

Code:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Dataset
months = np.array([1, 2, 3, 4, 5, 6]).reshape(-1, 1)
sales = np.array([100, 120, 130, 150, 170, 190])

# Train model
model = LinearRegression()
model.fit(months, sales)

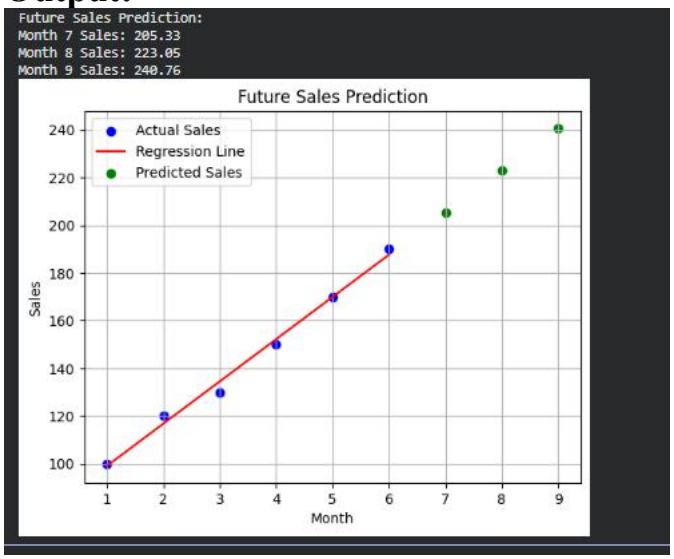
# Predict future sales
future_months = np.array([7, 8, 9]).reshape(-1, 1)
predicted_sales = model.predict(future_months)

print("Future Sales Prediction:")
for month, sale in zip(future_months.flatten(), predicted_sales):
    print("Month", month, "Sales:", round(sale, 2))

# Plot graph
plt.scatter(months, sales, color='blue', label='Actual Sales')
plt.plot(months, model.predict(months),
         color='red', label='Regression Line')
plt.scatter(future_months, predicted_sales,
           color='green', label='Predicted Sales')

plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Future Sales Prediction")
plt.legend()
plt.grid(True)
plt.show()
```

Output:


Future Sales Prediction:
Month 7 Sales: 205.33
Month 8 Sales: 223.05
Month 9 Sales: 240.76
Future Sales Prediction
Actual Sales
Regression Line
Predicted Sales
Sales
Month
240
220
200
180
160
140
120
100
1
2
3
4
5
6
7
8
9